

Optimal design of electrode placement in electro-tactile display and cutaneous sensory presentation to multiple fingers

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Figure 1: Example applications. (a) Changing shape recognition (e.g., converting the tactile perception of a cylindrical object into that of a square cylinder by presenting a sense of edges.); (b) Reproducing skin sensations lost from real objects, even when wearing finger sacks, to compensate for tactile sensations in real environments; (c) Tactile presentation that overlaps with the perceived skin sensation.

Abstract

When we manipulate objects, we use not only the finger pad, but also the entire fingertip, including the sides and hemispherical tip of the finger. Therefore, it is important to understand the tactile acuity distribution of the entire fingertip in order to realize a more natural and intuitive tactile experience. We measured the distribution of tactile acuity at the entire fingertip in detail and found that the tactile acuity changes almost linearly from the tip to the finger pad, and that the resolution decreases steeply at the side of the finger. Based on these measurements, we designed an optimal electrode for the electro-tactile device. Specifically, the range of sensory presentation was extended from the DIP-joint to the tip of the finger, the spacing between electrodes was linearly narrowed from the joint side to the tip, and the spacing between electrodes on the side of the finger was significantly increased compared to the finger pad area. This optimization enables a wider and higher density tactile presentation compared to the conventional one and is expected to increase the speed and accuracy of recognition and manipulation tasks using tactile sensation by increasing the amount of information transmitted.

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CCS Concepts

• **Human-centered computing** → **Haptic devices**; *Virtual reality*; *Mixed / augmented reality*.

Keywords

haptics, tactile acuity, electro-tactile, fingertips

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1 Introduction

When we manipulate an object, we immediately perceive its shape and surface properties through tactile sensations. This tactile information serves as feedback to fine-tune grip and manipulate objects effectively. For instance, when holding a wet glass, grip force is adjusted to prevent slipping, and when holding a spherical object, the placement of the fingers is optimized. These actions occur unconsciously and demonstrate highly precise and adaptable responses. Consequently, the sense of touch plays an important role in human motor control, and numerous studies have been conducted in various fields, including virtual reality (VR) and robotics, on techniques for presenting tactile feedback.

However, in most of these studies, only the region of the finger pad (in this paper, the term 'finger pad' refers to the flat area of

the fingertip) was used as the tactile presentation for the fingertip (The term 'fingertip' refers to the entire terminal segment of the finger in this paper). In contrast, when we manipulate an object (e.g., handling a pen or tool), we use not only the finger pad but also the side of the finger and the hemispherical part at the fingertip's end to distribute contact pressure and achieve a stable grip. This suggests that to achieve a more natural and intuitive tactile experience that closely mimics real-world object manipulation, it is necessary to present tactile sensations across a wider area of the entire fingertip, not just the finger pad.

To design tactile feedback across the entire fingertip, it is necessary to understand the distribution of tactile acuity across this region. However, specific changes in the distribution of tactile acuity at the fingertip, particularly on the sides of the finger, remain unclear. Therefore, we measured the tactile acuity distribution of the entire fingertip, including the finger pad, the fingertip end, and the lateral surface, for the index finger [1] and thumb.

The results reaffirmed a previously known finding: the resolution at the fingertip end is higher than that at the finger pad and also revealed that the change in resolution is almost linear. Additionally, we found that the tactile acuity on the sides of the index finger and thumb is approximately half as good as that of the finger pad.

In this exhibit, we will demonstrate a haptic device featuring an electro-tactile display, developed based on the measured fingertip tactile acuity distribution (Figure 1). The distance between the electro-tactile display electrodes was adjusted according to fingertip location, and the stimulation area was expanded to cover the entire fingertip. Additionally, the electrode layout was optimally designed to enable high-density tactile feedback over a wide area. This design is expected to enhance the richness of tactile information transmitted, thereby enabling rapid perception of an object's shape.

2 Related Work

2.1 Measuring the distribution of tactile acuity at the fingertips

To design a "whole-fingertip tactile presentation," it is essential to understand the distribution of tactile acuity across the fingertip. Previous studies have shown differences in tactile acuity between the flat and hemispherical portions of the fingertip (Johansson et al. [2]; Vallbo et al. [3]) and higher receptor density in the terminal hemisphere (Corniani et al. [4]). However, changes in tactile acuity distribution between the finger pad and the fingertip end, as well as the resolution of the finger's side, remain unclear.

Traditionally, tactile acuity has been assessed using the two-point discrimination threshold, which involves stimulating two points [5] [6] [7]. However, this method has been noted to have limitations, particularly due to variability in participants' responses, making it less suitable for accurately measuring tactile acuity [8]. An alternative method, two-point orientation discrimination (2POD), was proposed [9], but challenges in controlling force, pressure, and duration have limited its application. Instead, pressing a plate with parallel grooves against the skin and having participants identify its orientation has become the standard method [10] [11] [12]. Some studies have also used a pen-shaped stimulator with a semicircular contactor to measure direction discrimination [13].

However, these methods are unsuitable for measuring tactile acuity distribution in small areas, as required in this study. To address this, we developed a new method combining the classical two-point discrimination technique with the stripe pattern orientation method. Our approach uses a short linear contactor to determine the direction of a stripe pattern. Using this method and the method of constant stimuli, we measured the tactile acuity distribution across the index finger [1] and thumb. Details are provided in Chapter 3.

2.2 Extensive and dense tactile presentation at the fingertips

Tactile presentation technologies have been widely studied in fields such as virtual reality (VR) and robotics, with the goal of enhancing user immersion and improving object manipulation. Human fingertips play a crucial role in perceiving and handling objects by detecting their shape and surface properties. Various methods for delivering tactile sensations to the fingertips have been proposed, including wire traction [14], braking mechanisms for force feedback [15], pressure-induced force illusions [16], and vibratory feedback [17]. However, these techniques typically provide a single type of feedback (e.g., force or vibration) to a single finger. This limitation makes them unsuitable for high-density tactile feedback, which is necessary to perceive detailed features such as the edge of an object through the skin. For example, when handling a pencil, we rely on tactile sensations to detect its edges, enabling us to hold and align it precisely. Achieving this level of dexterity requires high-density tactile feedback that current methods cannot adequately provide.

To address this challenge, high-density tactile presentation techniques such as mechanical pin arrays [18], air pressure systems [19], suction devices [20], and electro-osmosis pads [21] have been developed. While these methods have advanced the field, they are largely limited to delivering feedback to the flat area of the finger pad. In contrast, electro-tactile stimulation has emerged as a promising alternative. It offers a compact and cost-effective solution capable of providing wide-range, high-density tactile feedback, including the curved surfaces of the fingertip. Recent innovations include lightweight electro-tactile displays [22], ultra-thin skin-like substrates [23], and perforated structures for tactile AR applications [24]. Despite these advances, current systems still primarily focus on the finger pad, leaving other fingertip areas underexplored.

To support grasping motions, tactile presentation to multiple fingers without restricting movement is essential. A method of applying electro-tactile stimuli to multiple fingers without mechanical components has been proposed [25] [26]. However, as tactile acuity differs across fingers, the electrode spacing must be tailored for each finger. Furthermore, excessive stimulation points can cause discomfort, necessitating the optimization of electrode placement. Although alternative electrode designs for each finger have been proposed [27], these designs lack physiological validation, highlighting the need for further research.

3 Measuring the Distribution of Tactile Acuity at the Fingertips

This chapter provides an overview of the methodology used to measure the tactile acuity distribution across the fingertips. For a

detailed explanation of the methodology and results for the index finger, please refer to our previous report [1].

3.1 Experiments

The experimental setup is shown in Figure 2. Contactors shaped like flat-blade screwdrivers were used in the experiment. The tactile acuity of the participant was measured by pressing the contactor against the fingertip and instructing the participant to indicate the direction of the contactor's edge. The contactors were applied to the skin either longitudinally or transversally, and participants were instructed to identify the direction of the contactor.

The contactors were fabricated using an optical 3D printer (Formlabs, Form3) and featured tips with a width (thickness) of 0.5 mm and edge lengths of 1.0, 1.5, 2.0, and 3.0 mm. A spring mechanism with a spring constant of 0.40 N/mm was installed at the tip (Figure 3), allowing for force adjustment. The force applied to the fingertip was set to 15 g by adjusting the spring's compression depth. Two control methods are available for the pressing mechanism: displacement control and force control. In this study, force control was employed using the spring mechanism. This method enables the experimenter to easily regulate the applied force by manually pressing the contactor.

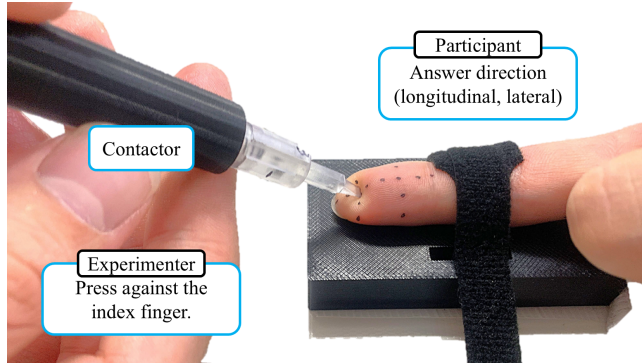


Figure 2: Experimental setup

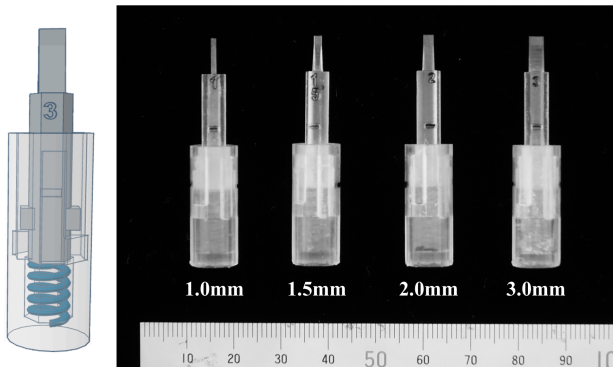


Figure 3: Tip of the fabricated contactors (left: 3D model, right: actual contactors).

3.2 Results

The blue line of Figure 4 represents the 75% threshold at each position in the row from the tip of the index fingertip to the joint side. The tactile acuity distribution measured in this study differs from the conventional two-point discrimination threshold in terms of measurement methodology. Therefore, it cannot be stated that the obtained values represent the tactile spatial resolution in a traditional sense. On the other hand, previous studies have commonly reported two-point discrimination thresholds of 2 mm or 3 mm for the finger pad. The results of the present study are closely aligned with these values, albeit slightly smaller.

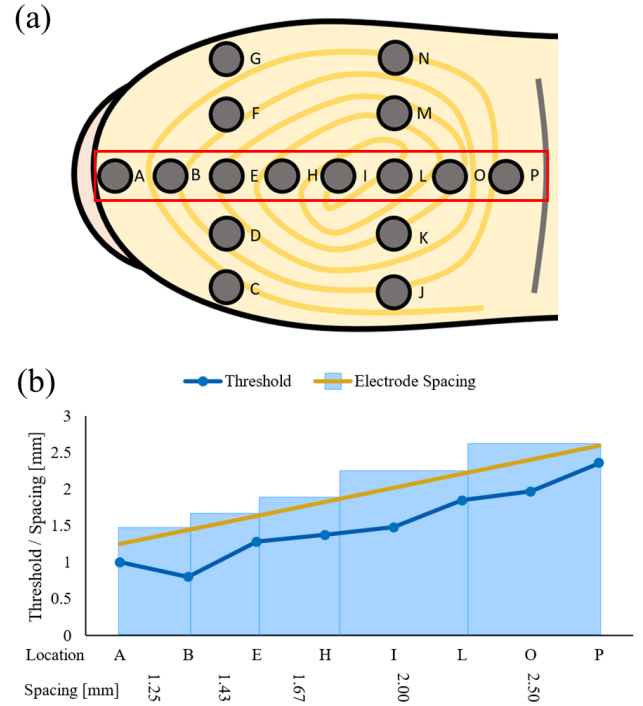


Figure 4: (a) Simulation point; (b) Comparison of distribution of tactile acuity and designed electrode spacing of the index finger.

4 Development of Electro-tactile Display

4.1 Device

The electro-tactile display used in this study is described as follows: Figure 5 (a) depicts the appearance of the device, while Figure 6 illustrates the system configuration. The device connects to a PC via USB and is powered through USB bus power. It internally generates a voltage of 300 V to produce a current-controlled electro-tactile stimulation. The stimulation pulses are distributed to the electrodes through a switching board. Each board supports 64 stimulation channels, and by serially connecting two boards, the number of stimulating electrodes is expanded to 128. The intensity of the electro-tactile stimulation can be adjusted at any time using a keyboard interface. Electro-tactile stimulation is delivered at a

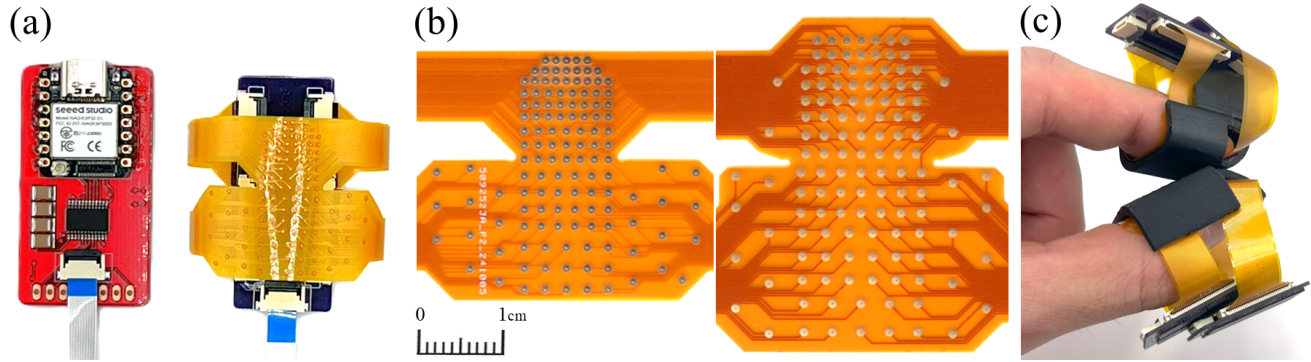


Figure 5: Device and Electrodes. (a) Appearance of the device (left: main circuit board, right: flexible circuit board for electrical stimulation at the index finger); (b) Created Electrodes (left: Index finger, right: Thumb); (c) Attached view.

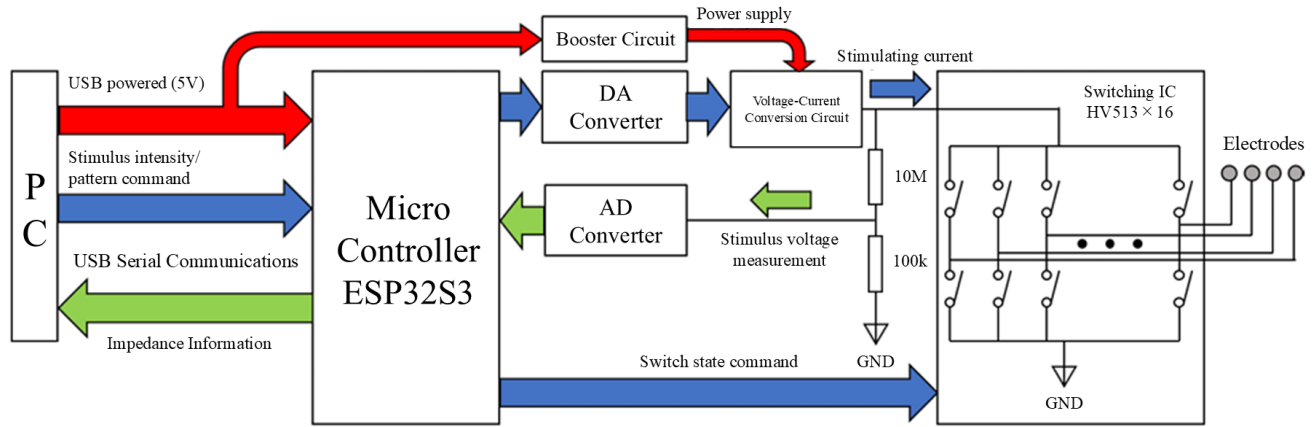


Figure 6: System configuration

frequency of 90 pulses per second for each active electrode. Each pulse has a width of 50 μ s, and the maximum pulse current is 10 mA.

4.2 Electrodes Design

The electrode spacing was designed based on the measured tactile acuity distribution (Figure 5 (b)). To extend the stimulation range across the entire fingertip, including the sides and tip, the electrode spacing was linearly reduced from the joint side of the fingertip toward the tip, as shown in Figure 4. Additionally, the spacing between electrodes on the sides of the fingertip was made significantly wider than that on the finger pad.

Each electrode was 1 mm in diameter, and the flexible circuit board was designed with a front-and-back slit structure to conform to the curved shape of the fingertip. The arrangement of electrodes, as shown in Figure 4, achieved tactile acuity slightly exceeding the measured resolution. However, the number of stimulation points was constrained to avoid excessive circuit complexity. A significant increase in electrode numbers would require additional switching

circuits, leading to higher stray capacitance, which makes it difficult to perform electro-tactile stimulation.

This design represents the first implementation capable of covering the entire fingertip with stimulation points, enabling tactile presentations that extend to multiple fingers.

5 Example Applications

The device offers wide-range, high-density tactile presentation across the entire fingertip, utilizing an electrode array optimized for each fingertip region. This design enables not only the perception of surface characteristics, textures, and vibrations, as provided by conventional electro-tactile displays, but also clearer recognition of the shape and surface texture of virtual objects.

For instance, the device can be used for haptic augmented reality (AR), where tactile information is overlaid on real objects. One example is presenting edge sensations on a cylindrical object to make it feel like a rectangular prism. In addition, combined with film-type tactile sensors, it is possible to enhance skin sensations from real objects. This feature is expected to serve as a novel method for supplementing tactile perception in real-world environments.

Furthermore, when used without finger sacks and equipped with flexible electrodes, the device allows tactile presentations to seamlessly overlap with natural skin sensations, ensuring a natural and intuitive operational feel.

In this exhibit, we demonstrate a shape recognition task using two fingers, the thumb and index finger, in combination with the TouchX force-sensing presentation device (Figure 1). The TouchX is utilized as a position and posture sensor in this setup, and its force presentation function is not employed. A real object, such as a cylinder, is placed on a desk, and the user interacts with it. During this interaction, the real object is perceived as a rectangular cylinder due to the electro-tactile sensation provided by the device. This demonstration shows not only a novel form of haptic AR but also the potential of wide-range, high-precision haptic feedback for teleoperation. By presenting high-resolution tactile sensations, detailed 3D shape information can be effectively transmitted.

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